

MATH 4175 - Project 5

AES-128 First Round Encryption

Group Members: Luke Freeman, Yoshwan Pathipati, Diego Penadillo, Hiren Sai Vellanki

1. Input

Plaintext message: "I need an A+ plz"

Plaintext (hex): 49 20 6E 65 65 64 20 61 6E 20 41 2B 20 70 6C 7A

AES Key (hex): 45 6E 63 72 79 70 74 20 6D 79 20 6D 61 72 6B 73

2. Round Key K0

The initial round key K0 is the AES key bytes arranged column-wise into a 4x4 state matrix.

K0 Matrix

```
[45 79 6D 61]
[6E 70 79 72]
[63 74 20 6B]
[72 20 6D 73]
```

3. Round Key K1 (Key Schedule)

The AES-128 key schedule generates round keys by expanding the original 16-byte key into 44 words (4 bytes each).

Key Schedule Steps for Round 1

Initial words from the key:

$W_0 = [45\ 6E\ 63\ 72]$

$W_1 = [79\ 70\ 74\ 20]$

$W_2 = [6D\ 79\ 20\ 6D]$

$W_3 = [61\ 72\ 6B\ 73]$

Computing W4:

$\text{RotWord}(W_3) = [72\ 6B\ 73\ 61]$

$\text{SubWord}(\dots) = [40\ 7F\ 8F\ EF]$

$\text{XOR with Rcon}[0] = [01\ 00\ 00\ 00]$

$g(W_3) = [41\ 7F\ 8F\ EF]$

$W_4 = W_0 \text{ XOR } g(W_3) = [04\ 11\ EC\ 9D]$

$W_5 = W_1 \text{ XOR } W_4 = [7D\ 61\ 98\ BD]$

$W_6 = W_2 \text{ XOR } W_5 = [10\ 18\ B8\ D0]$

$W_7 = W_3 \text{ XOR } W_6 = [71\ 6A\ D3\ A3]$

K1 Matrix

```
[04 7D 10 71]
[11 61 18 6A]
[EC 98 B8 D3]
[9D BD D0 A3]
```

4. First Round Encryption

Initial State

The plaintext bytes are arranged column-wise into the 4x4 state matrix.

```
[49 65 6E 20]
[20 64 20 70]
[6E 20 41 6C]
[65 61 2B 7A]
```

(a) After AddRoundKey with K0

XOR each byte of the initial state with the corresponding byte of K0.

```
[0C 1C 03 41]
[4E 14 59 02]
[0D 54 61 07]
[17 41 46 09]
```

(b) After SubBytes

Apply the AES S-Box substitution to every byte in the state.

```
[FE 9C 7B 83]
[2F FA CB 77]
[D7 20 EF C5]
[F0 83 5A 01]
```

(c) After ShiftRows

Cyclically shift each row to the left: Row 0 no shift, Row 1 shift left by 1, Row 2 shift left by 2, Row 3 shift left by 3.

```
[FE 9C 7B 83]
[FA CB 77 2F]
[EF C5 D7 20]
[01 F0 83 5A]
```

(d) After MixColumns

Apply the MixColumns transformation using GF(2⁸) multiplication with irreducible polynomial $x^8 + x^4 + x^3 + x + 1$.

```
[02 03 01 01]
[01 02 03 01]
[01 01 02 03]
```

[03 01 01 02]

[1C 50 3B 16]

[3A B5 74 E7]

[C2 CD 27 02]

[0E 4A 30 25]

(e) After AddRoundKey with K1

XOR each byte of the state with the corresponding byte of K1. This is the final state after the first AES round.

[18 2D 2B 67]

[2B D4 6C 8D]

[2E 55 9F D1]

[93 F7 E0 86]

5. Explanation of Each Step

1. AddRoundKey (K0): The plaintext state is XORed with the initial round key K0. This ensures the encryption is key-dependent from the very first operation.

2. SubBytes: A non-linear byte substitution using the AES S-Box. This provides confusion, making the relationship between the key and ciphertext complex.

3. ShiftRows: Each row is cyclically shifted left by a different amount. This provides diffusion by permuting bytes across columns.

4. MixColumns: Each column is multiplied by a fixed polynomial matrix over $GF(2^8)$. This provides further diffusion by mixing bytes within each column.

5. AddRoundKey (K1): The state is XORed with the first expanded round key K1. This ensures the result is key-dependent before proceeding to subsequent rounds.

6. Validation

The Python script was executed successfully and all results were independently verified.

All matrices are 4x4: Yes. Each state matrix has exactly 4 rows and 4 columns.

All entries are two-digit uppercase hex: Yes. Every byte is formatted as 02X.

K0 matches the original key arranged column-wise: Yes. K0 is the key bytes [45 6E 63 72 79 70 74 20 6D 79 20 6D 61 72 6B 73] placed column by column.

K1 matches the AES-128 key schedule: Yes. K1 was computed using RotWord, SubWord, Rcon, and XOR as specified by the AES standard.

Final first-round state (after AddRoundKey with K1):

[18 2D 2B 67]

[2B D4 6C 8D]

[2E 55 9F D1]

[93 F7 E0 86]

Source code: `project5_aes_first_round.py` | Generated programmatically to ensure accuracy.